Unit: Unit 0: Foundations of Algebra
Topic: Divisibility Division Algorithm

Name: ____________________
Date: ____________

Foundation (Jalapeno)

Q1. Which of the following numbers is divisible by 2 and 3?
(A) 7
(B) 10
(C) 12
(D) 15
(E) 21

Q2. If a stock market investor has 45 shares to sell, how many complete sets of 9 shares can be sold?
(A) 4
(B) 5
(C) 6
(D) 7
(E) 8

Q3. Determine if 75 is divisible by 5.
(A) Yes
(B) No
(C) Maybe
(D) Cannot determine
(E) Insufficient information

Q4. A savings account has 240 dollars. If this amount is divided into sets of 10, how many complete sets can be made?
(A) 20
(B) 22
(C) 24
(D) 26
(E) 28

Q5. If 18 is divided by 3, what is the remainder?
(A) 0
(B) 1
(C) 2
(D) 3
(E) 4

Q6. How many times does 9 go into 63 with a remainder of 0?
(A) 5
(B) 6
(C) 7
(D) 8
(E) 9

Q7. Determine the divisibility of 48 by 2 and 8.
(A) Divisible by 2 only
(B) Divisible by 8 only
(C) Divisible by both 2 and 8
(D) Not divisible by either
(E) Cannot determine

Q8. A tax amount of 150 dollars is to be divided among 5 people. How much will each person pay?
(A) 25
(B) 30
(C) 35
(D) 40
(E) 45

Open Ended Questions (FRQ)

Q9. Perform the long division to find the quotient and remainder when 48 is divided by 7. Express the result in the form $dividend = quotient \cdot divisor + remainder$.

Q10. If a stock market investor buys 36 shares of stock, and wants to put them into boxes that hold 4 shares each, how many boxes can be completely filled? Show your work and explain your reasoning.

Chef's Tips

- Tip 1: Ensure students understand the concept of divisibility before moving to division algorithms
- Tip 2: Provide numerous examples and word problems to reinforce understanding
- Tip 3: Encourage students to double-check calculations for accuracy